

NEET 2018

Set NN

Solution – Biology

46. The correct option is 3.

Parathyroid hormone and estrogen plays an important role in osteoporosis. Parathyroid hormone maintains the calcium level by stimulating both resorption and formation of bone. Estrogen inhibit bone resorption.

47. The correct option is 2.

Epinephrine is an amino acid derivative hormone. Estradiol, ecdysone and estriol are steroidal in nature.

48. The correct option is 4.

49. The correct option is 2.

Eye lens is held in place by suspending ligaments to ciliary muscles.

50. The correct option is 2.

The amnion arises from **extra-embryonic** somatic mesoderm on the outer side and the **extra-embryonic** ectoderm on the inner side.

51. The correct option is 3.

Human chorionic gonadotropin (HCG), estrogen, progesterone and human placental lactogen (hPL) are the 4 major hormones secreted by placenta.

52. The correct option is 2.

SAHELI is a non-hormonal contraceptive birth control pill. It prevents the implantation by blocking the estrogen receptors in the uterus.

53. The correct option is 1.

In spermiogenesis, spermatids form motile spermatozoa. Spermiation is a process in which mature spermatids are released from Sertoli cells into the seminiferous tubules.

54. The correct option is 1.

As mother has X linked condition on 1 chromosome, both sons and daughter will inherit. Son will be diseased and daughters will be carrier if the father is normal.

55. The correct option is 4.

According to Hugo de Vries, the mechanism of evolution is saltation or single step large mutation.

56. The correct option is 3.

Enhancers are not a part of operon. It is a short region of DNA that can be bound by proteins (activators) to increase the likelihood that transcription of a particular gene will occur.

57. The correct option is 1.

58. The correct option is 3.

Proliferative phase is also known as follicular phase. Breakdown of endometrial lining is menstruation, and secretory phase is luteal phase.

59. The correct option is 2.

Elephantiasis or filariasis is the mosquito transmitted disease that causes chronic inflammation to lymphatic vessels.

60. The correct option is 1.

Eye of octopus, mammal are formed due to convergent evolution.

61. The correct option is 2.

Homologous structures have same ancestors but the structures will have different functions.

62. The correct option is 3.

Alzheimer's disease is not an autoimmune disease.

63. The correct option is 2.

Inheritance of blood group is an example of co-dominance, polygenic inheritance and multiple alleles.

64. The correct option is 3.

Vitamin B12 content is increased when milk is converted into curd.

65. The correct option is 3.

Eutrophication is nutrient enrichment. Sanitary landfill is associated with waste disposal. UV-B radiation causes snow blindness. Jhum cultivation is associated with deforestation.

66. The correct option is 1.

67. The correct option is 4.

Sacred grooves are an example of in-situ conservation.

68. The correct option is 1.

For a growing population, pre-reproductive individuals are always less than the reproductive individuals.

69. The correct option is 2.

Poppy flowers produce a milky fluid which works as a source of raw opium. This will produce smack.

70. The correct option is 1.

Parietal cells produce intrinsic factor which is necessary for the absorption of Vitamin B12. Inability to absorb vitamin B12 is associated with ineffective erythropoiesis.

71. The correct option is 1.

Fibrinogen is associated with blood clotting, albumins helps in osmotic balance and globulin are part of defence mechanism.

72. The option is 4.

Silicosis is an occupational lung disorder. Inhalation of silica dust causes silicosis.

73. The correct option is 2.

Calcium released from sarcoplasmic reticulum causes a displacement of troponin from the active sites on actin where the myosin will bind to form cross bridges for muscle contraction.

74. The correct option is 1.

Nissl bodies are composed of rough ER and free ribosomes.

75. The correct option is 1.

Oxidative phosphorylation occurs in inner mitochondrial membrane and not in outer mitochondrial membrane.

76. The correct option is 1.

Lampbrush chromosomes are found in amphibian oocytes. Polytene chromosomes are found in larvae of some dipterans.

77. The correct option is 4.

Humans teeth are diphyodont, heterodont and thecodont.

78. The correct option is 1.

Phospholipid synthesis occur in smooth ER.

79. The correct option is 2.

Polysomes or polyribosomes are many ribosomes associated with single mRNA to form multiple copies of polypeptide simultaneously.

80. The correct option is 2.

81. The correct option is 3.

Crop and gizzard are found in birds. They are the part of the alimentary canal.

82. The correct option is 4.

Chelone is basically a plant. Rest of them are animals and are homeotherm.

83. The correct option is 4.

Males have caudal or anal styles whereas female cockroach have anal cerci.

84. The correct option is 2.

Earthworm do not undergo metamorphosis.

85. The correct option is 4.

Diatoms are the chief producers in the oceans. They are a type of phytoplankton.

86. The correct option is 1

87. The correct option is 2.

Tricuspid valve is present between the right atrium and right ventricle. Bicuspid valve is present between the left ventricle and left atrium. Semilunar valves are present between right ventricle and pulmonary artery.

88. The correct option is 4.

Tidal volume- 500 to 550 ml

Inspiratory reserve volume- 2100-3200 ml

Expiratory reserve volume- 1000 to 1200 ml

Residual volume- around 1200 ml

89. The correct option is 1.

Glycosuria- presence of glucose in urine

Gout- accumulation of uric acid in joints

Renal calculi- mass of crystallized salts within the kidney

Glomerular nephritis- inflammation of glomeruli

90. The correct option is 4.

Ultrafiltration occurs in glomerulus/Malpighian corpuscle. Transport of urine occurs in ureter.

Storage of the urine occurs in urinary bladder. Concentration of urine occurs in Henle's loop.

91. The correct option is 1.

92. The correct option is 2

Halophytes contain pneumatophores. They are also known as breathing roots. It helps in gaseous exchange.

93. The correct option is 2.

Grasses are monocots, do not have secondary growth

94.

95. The correct option is 1.

Casparian strips occurs in endodermis. It is an inner layer of cells in the cortex of a root and of some stems, surrounding a vascular bundle.

96. The correct option is 3.

Sweet potato is an example of tuberous root. It is a modified tap root.

97.

98. The correct option is 2.

The pyramid is inverted pyramid of biomass.

99. The correct option is 3.

16 September is celebrated as world ozone day.

100. The correct option is 4.

Cl acts as catalyst for degradation of ozone. It is release due to chlorofluorocarbons.

101. The correct option is 4.

Natality is birth rate whereas mortality is death rate.

102. The correct option is 1.

The functional role that is played by an organism where it lives is known as niche.

103. The correct option is 1.

Ozone is a secondary pollutant.

104. the correct option is 1.

Winged pollen grains are present in *Pinus*.

105. The correct option is 3.

Agaricus spores are produced exogenously.

106.

107. The correct option is 1.

108. The correct option is 4.

Yucca has a very close relationship with moth. They cannot complete their life cycle without each other.

109. The correct option is 3.

Liquid nitrogen is used to store pollen grains at -196°C

110. The correct option is 4.

Ferrous is the form which is absorbed by the plants.

111. The correct option is 3.

Potassium maintains the turgidity of the plant cells. They also regulate opening and closing of stomata.

112. The correct option is 1.

Syngamy and triple fusion occurs during double fertilization. Triple fusion forms endosperm

113. The correct option is 3

114. The correct option is 2.

Green sulphur bacteria do not produce oxygen during photosynthesis.

115. The correct option is 4.

Secretory vesicles bud from Golgi and carry transport proteins to target cell organelles.

116. The correct option is 3.

Oxygen concentration does not determine the opening or closing of stomata. It is the temperature, light and carbon-dioxide concentration that determines the opening of stomata.

117. The correct option is 2.

Grasses are monocots, so the shape of the stomata should be dumb-bell shaped.

118. The correct option is 1.

Site for ribosomal RNA synthesis.

119. The correct option is 4.

NADH is not the end product of light reactions. ATP, NADPH, oxygen are formed at the end of light reactions.

120. The correct option is 2.

Separation of homologous chromosomes begins in pachytene.

121. The correct option is 1.

Carbonyl and hydroxyl are two functional groups found in carbohydrates.

122. The correct option is 2.

Saccharomyces, is a unicellular prokaryote.

123. The correct option is 4.

Mitotic divisions/vegetative propagation produce the offsets.

124. The correct option is 4.

Punnett square was developed by British scientist name as Reginald **Punnett** and William Bateson

125. The correct option is 1.

Sporopollenin is found in the outer layer of pollen and is preserved in soil and sediments.

126. The correct option is 2.

Starch synthesis in pea is not an example of multiple alleles.

127. The correct option is 1

The Discovery of the lac Operon. In **1961**, Francois Jacob and Jacques Monod proposed the operon model of gene regulation in bacteria. The model was based on their study of the genes in *E. coli* that code for enzymes that affect the breakdown of lactose. Francois Jacob and Jacques Monod.

128. The correct option is 2.

Most bamboo plants flower once in a life cycle of about 40 to 50 years.

129. The correct option is 4.

The proof for semi-conservative replication was conducted on *E.coli*, which is a bacterium.

130. The correct option is 2.

Retroviruses are used for introducing DNA fragments in human lymphocytes.

131. The correct option is 2.

Ribozyme are nucleic acid molecules that are capable of catalysing different reactions.

132. The correct option is 4.

Biopiracy is defined as the practice of commercially exploiting naturally occurring biochemical or genetic material, especially by obtaining patents that restrict its future use, while failing to pay fair compensation to the community from which it originates.

133. The correct option is 1.

Basmati patent case is won by India but the trademark issue remains.

134. The correct option is 1.

GEAC is involved in assessing the safety of introducing genetically modified organisms for public use.

135. The correct option is 1.

Denaturation is the first step of PCR, the annealing and finally extension.